Week 4 | **CLO** 2-A · 3-C · 2-D (R) | **Before** In-Class Quiz I | **File** 4 / 8

3-3 Measures of Position

مقاييس الموقع — الدرجة المعيارية والمئينات والربيعات والعشيرات والقيم الشاذة

Measures of position (location) give the **relative position of a data value** compared with the other values in a data set: **standard scores (z)**, **percentiles**, **deciles** and **quartiles**. BOTH

1 · Standard scores (z-scores) HADHRAMI

Z-SCORE: HOW MANY SDS A VALUE LIES ABOVE (+) OR BELOW (−) THE MEAN

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

$$\text{sample: } z = \frac{X - \bar{x}}{s}$$

$$\text{population: } z = \frac{X - \mu}{\sigma}$$

$z = 0$: the value equals the mean. Rearranged: $X = \bar{x} + z \cdot s$. If *every* value is converted to z, the new data have **mean 0 and SD 1**.

EX 3-27 · TEST SCORES

English 85 (mean 76, s = 4) and French 42 (mean 36, s = 3). Compare her relative positions.

$z_E = (85 - 76)/4 =$ **2.25** $\quad z_F = (42 - 36)/3 =$ **2.00**

Her relative position is higher in **English**.

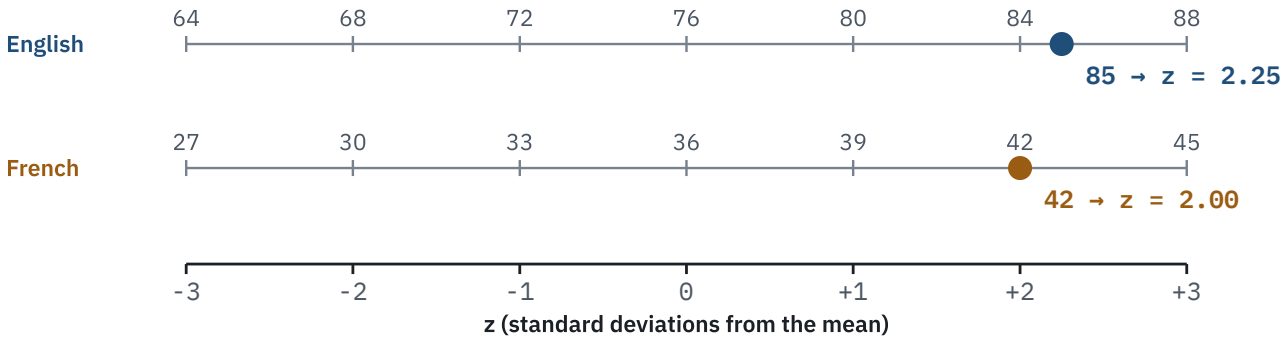
EX 3-28 · MARRIAGE AGES

Men: mean 26.4, s = 2; a man marries at 24. Women: mean 23.5, s = 2.3; a woman marries at 22.

$z_{man} = (24 - 26.4)/2 =$ **−1.2**

$z_{woman} = (22 - 23.5)/2.3 \approx$ **−0.65**

The woman's age at marriage is relatively **higher**: −0.65 is closer to her group's mean than −1.2 is to his.



Example 3-27. Raw scores 85 and 42 cannot be compared directly. On the common z scale, English (z = 2.25) sits further above its class mean than French (z = 2.00), so English is her better relative position.

مثال 27-3: لا يمكن مقارنة 85 و 42 مباشرة. على مقياس z الموحد، الإنجليزي (z = 2.25) أبعد فوق متوسط فصله من الفرنسي (z = 2.00)، إذن موقعها النسبي في الإنجليزي أفضل.

الشرح بالعربي — الدرجة المعيارية z

تقيس **كم انحرافًا معياريًا** تبعد القيمة عن المتوسط: `z = (X − x̄)/s` . الموجبة تعني فوق المتوسط، والسالبة تحته، والصفر يعني عند المتوسط تمامًا. فائدتها: **مقارنة قيم من مجموعات مختلفة**. درجة 85 في الإنجليزي لا تُقارن مباشرة بـ 42 في الفرنسي، لكن `z = 2.25` أفضل من `z = 2.00` . انتبه مع الأرقام السالبة: `−0.65` أكبر من `−1.2` ، أي أفضل موقعًا.

2 · Percentiles BOTH

Percentiles divide **ordered** data into 100 equal parts, written $P_1, P_2, \ldots, P_{99}$. The p th percentile P_p is the value that is greater than p% of the data values: p% lie below it and (100 − p)% above. P_1: 1% below. P_{50} is the **median**. P_{99}: 99% below. There is **no 100th percentile**: a percentile counts the data *below* a value, and no value can be higher than the highest value in the data set, so percentiles run from 1 to 99. They are used in education and health (growth charts).

You are the 4th tallest of 20. 16 people are shorter than you: 16/20 × 100 = 80%, so you are at the **80th percentile, P_{80}**. If your height is 1.85 m, then 80% of the group are shorter than 1.85 m. AMANI

PERCENTILE ≠ PERCENTAGE AMANI

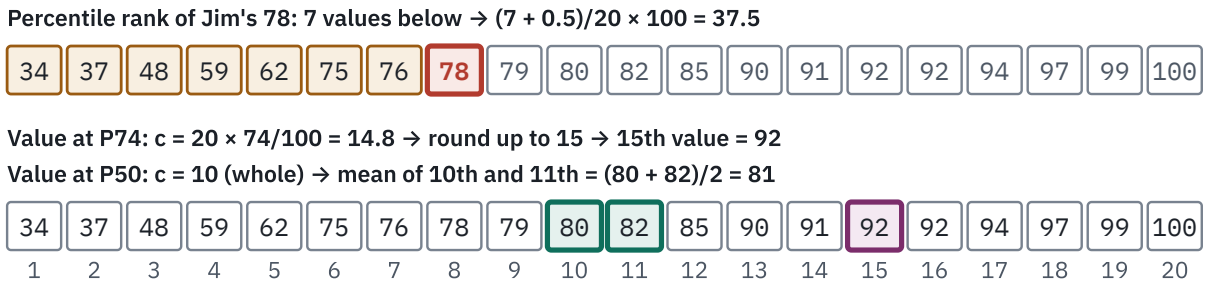
A **percentage** measures your own score (40 out of 50 = 80%) and compares you with nobody. A **percentile** measures your position relative to others.

If 100 people sit a 50-question test and you get 40 right (80%), you don't know your percentile until you see the other scores. If all 99 others scored 60% or less, you beat 99% of the group: **99th percentile**. It is quite possible to score 60% and be in the 95th percentile on a hard exam.

CASE 1 · PERCENTILE RANK OF A GIVEN VALUE X

PERCENTILE FORMULA (ORDER THE DATA FIRST)

$$\text{Percentile of } X = \frac{(\text{number of values below } X) + 0.5}{\text{total number of values}} \cdot 100$$



Top: count the values strictly below X, add 0.5, divide by n. Bottom: c = n·p/100 gives a position. A decimal c rounds up to one position; a whole c means averaging positions c and c + 1.

أعلى: عُدّ القيم الأقل من X، أضف 0.5، واقسم على n. أسفل: c = n·p/100 يعطي الترتيب؛ إذا كان c عشريًا نقرّبه لأعلى، وإذا كان عددًا صحيحًا نأخذ متوسط القيمتين c و c + 1.

ACTIVITY · JIM'S 78 AMANI

20 scores: 34 37 48 59 62 75 76 78 79 80 82 85 90 91 92 92 94 97 99 100.

7 values are below 78: `(7 + 0.5)/20 × 100 =` **37.5th percentile**. 37.5% of students scored lower than Jim.

EX 3-30, 3-31 · TRAFFIC VIOLATIONS HADHRAMI

10 days: 22 19 25 24 18 15 9 12 16 20. Ordered: 9 12 15 16 18 19 20 22 24 25. Percentile rank of 16, and of 24.

16: 3 values below, (3.5/10)·100 = **35th**. 24: 8 below, (8.5/10)·100 = **85th**. So 24 is higher than 85% of the values.

CASE 2 · THE VALUE AT A GIVEN PERCENTILE P

PROCEDURE: (1) ORDER THE DATA · (2) C = N·P/100

$$c = \frac{n \cdot p}{100} \quad (n = \text{number of values}, p = \text{percentile})$$

3A: if c is **not** a whole number, round **up** to the next whole number and count up to that position. **3B**: if c **is** a whole number, take the value halfway between the c-th and (c + 1)-th values.

ACTIVITY · JIM'S CLASS: P_{50} AND P_{74} AMANI

P_{50}: c = 20·50/100 = 10, a whole number, so `(10th + 11th)/2 = (80 + 82)/2 =` **81**.

P_{74}: c = 20·74/100 = 14.8, not whole, so round up to 15. The 15th value = **92**.

EX 3-32, 3-33 · TRAFFIC: P_{65} AND P_{30} HADHRAMI

P_{65}: c = 10·65/100 = 6.5, rounds up to 7. The 7th value = **20**.

P_{30}: c = 3, whole, so `(15 + 16)/2 =` **15.5**.

الشرح بالعربي — المئينات

المئين يخبرك **كم بالمئة من البيانات أقل من قيمتك**. مثلًا P_{80} يعني أن 80% أقل منك. لا تخلط بين **النسبة المئوية** (درجتك أنت من 100) و**المئين** (ترتيبك بين الآخرين). هناك سؤالان مختلفان: (1) **لديك قيمة وتريد ترتيبها** ← عدّ القيم الأقل منها، أضف 0.5، اقسم على n، واضرب في 100. (2) **لديك مئين وتريد قيمته** ← احسب $c = n \cdot p/100$. إذا كان c عشريًا قرّب **لأعلى** وخذ القيمة في ذلك الترتيب، وإذا كان صحيحًا خذ متوسط القيمتين c و c + 1. ولا تنسَ ترتيب البيانات أولًا.

3 · Quartiles BOTH

Quartiles divide ordered data into four equal groups: $Q_1 = P_{25}$, $Q_2 = P_{50} = \text{median}$, $Q_3 = P_{75}$.

Method 1, medians (the textbook procedure table): order the data; Q_2 = the median; Q_1 = the median of the values **below** Q_2; Q_3 = the median of the values **above** Q_2. When n is odd, leave the median itself out of both halves.

Method 2, percentile formula: apply the Case 2 procedure with p = 25, 50, 75.

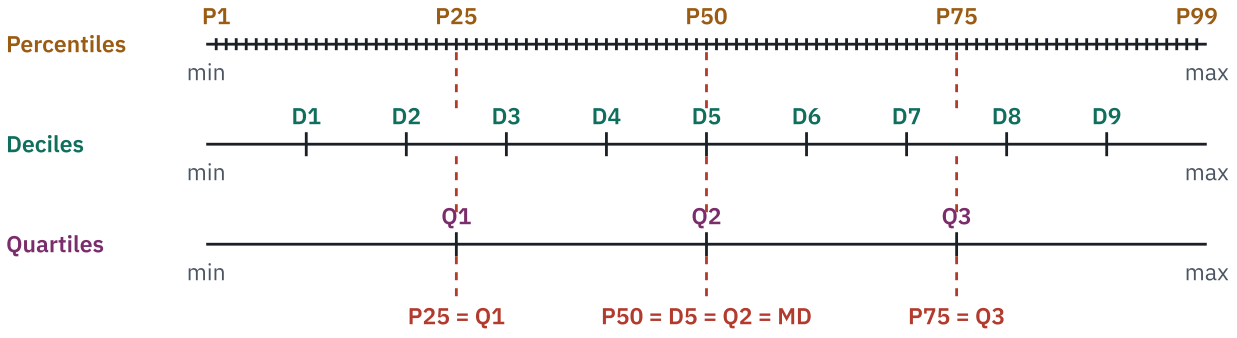
ACTIVITY · QUARTILES OF JIM'S CLASS, BOTH METHODS AMANI

Method 1: n = 20 (even), so Q_2 = (80 + 82)/2 = **81**. Lower ten: 34 … 80, Q_1 = (62 + 75)/2 = **68.5**. Upper ten: 82 … 100, Q_3 = (92 + 92)/2 = **92**.

Method 2: Q_1: c = 5, whole, so (5th + 6th)/2 = 68.5. Q_2: c = 10, giving 81. Q_3: c = 15, whole, so (15th + 16th)/2 = 92. Both methods agree.

EX 3-34 · TRAFFIC DATA (N = 10) HADHRAMI

MD = (18 + 19)/2 = **18.5**. Below: 9 12 **15** 16 18, so Q_1 = **15**. Above: 19 20 **22** 24 25, so Q_3 = **22**.



Percentiles cut ordered data into 100 parts, deciles into 10, quartiles into 4. They are the same scale: Q1 = P25, the median = P50 = D5 = Q2, Q3 = P75, and each Dk = P(10k).

المئينات تقسم البيانات المرتبة إلى 100 جزء، والعشيرات إلى 10، والربيعات إلى 4. وكلها على نفس المقياس: Q1 = P25، والوسيط = Q2 = D5 = P50، و Q3 = P75.

4 · Deciles and the summary of position measures HADHRAMI

Deciles divide the distribution into 10 groups: $D_1, \ldots, D_9$ correspond to $P_{10}, \ldots, P_{90}$, and are found with the percentile formulas. The median = $P_{50} = Q_2 = D_5$. Textbook Table 3-4 summarises all position measures.

5 · Interquartile range (IQR) BOTH

IQR: THE RANGE OF THE MIDDLE 50% OF THE DATA

$$IQR = Q_3 - Q_1$$

A rough measure of variability. Like the SD, the more variable the data, the larger the IQR. Unlike the SD, it hardly reacts to outliers.

EX 3-35 · TRAFFIC · JIM'S CLASS

Traffic: 22 − 15 = **7**. Jim's class: 92 − 68.5 = **23.5**.

6 · Outliers BOTH

An **outlier** is an extremely high or extremely low value compared with the rest of the data. Outliers strongly affect the **mean** and **standard deviation**; the **median** and **IQR** are much less affected.

PROCEDURE FOR IDENTIFYING OUTLIERS

(1) Order the data, find Q_1 and Q_3. (2) IQR = $Q_3 - Q_1$. (3) Compute 1.5 × IQR. (4) Lower fence Q_1 − 1.5·IQR; upper fence Q_3 + 1.5·IQR. (5) Any value **below** the lower fence or **above** the upper fence is an outlier.

Why outliers occur AMANI: (1) measurement or observational error; (2) recording error, such as a value written or typed wrongly; (3) a rare value that is real but occurred by chance.

EX 3-36 HADHRAMI

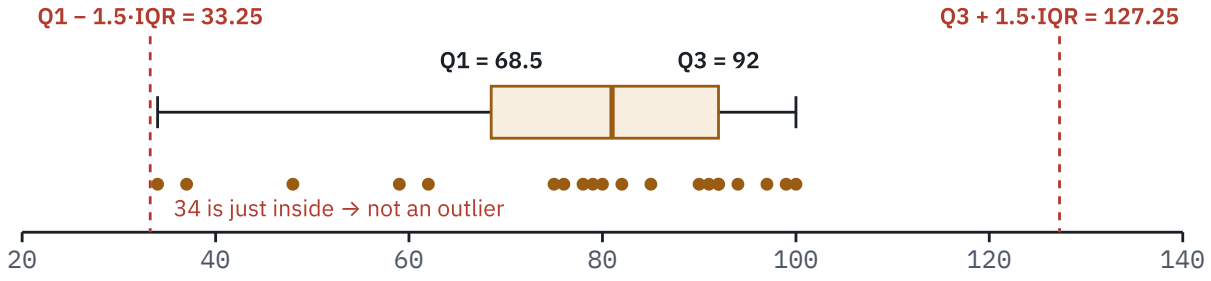
5, 6, 12, 13, 15, 18, 22, 50

$Q_1 = (6 + 12)/2 = 9$, $Q_3 = (18 + 22)/2 = 20$, IQR = 11, 1.5·IQR = 16.5.

Fences: 9 − 16.5 = −7.5 and 20 + 16.5 = 36.5. Since 50 > 36.5, **50 is an outlier**.

ACTIVITY · JIM'S CLASS AMANI

1.5 × 23.5 = 35.25. Fences: 68.5 − 35.25 = 33.25 and 92 + 35.25 = 127.25. The smallest value, 34, is above 33.25: **no outliers**.



IQR = 92 − 68.5 = 23.5, so 1.5·IQR = 35.25. Anything below 33.25 or above 127.25 would be an outlier. The lowest score, 34, misses the fence by 0.75, so there are no outliers.

المدى الربيعي = 23.5، و 1.5 × 23.5 = 35.25. أي قيمة أقل من 33.25 أو أكبر من 127.25 تعتبر قيمة شاذة. أقل درجة 34 داخل الحد بفارق 0.75 فقط، إذن لا توجد قيم شاذة.

الشرح بالعربي — الربيعات والقيم الشاذة

الربيعات تقسم البيانات المرتبة إلى **4 أجزاء**، Q_2 هو الوسيط، و Q_1 هو وسيط النصف الأدنى، و Q_3 هو وسيط النصف الأعلى. وإذا كان n فرديًا **لا ندخل الوسيط** في أي من النصفين. المدى الربيعي $IQR = Q_3 - Q_1$ هو مدى الـ 50% الوسطى. لكشف القيم الشاذة نحسب الحدّين $Q_1 - 1.5 \cdot IQR$ و $Q_3 + 1.5 \cdot IQR$ ، وأي قيمة **خارجهما** تعتبر شاذة. وانتبه: القيمة التي تساوي الحد تمامًا **ليست** شاذة. أسباب القيم الشاذة: خطأ في القياس، أو خطأ في التسجيل، أو قيمة نادرة حقيقية.

R FOR LAB QUIZ I (OUTLINE CLO2-D): KNOW WHY R CAN DISAGREE WITH YOUR PAPER ANSWER

`quantile(x, 0.25)`, `summary(x)` and `IQR(x)` use R's default method, which **interpolates** between positions. For Jim's 20 scores R gives $Q_1 = 71.75$ and $Q_3 = 92$ (IQR 20.25), not the 68.5 and 92 (IQR 23.5) from the procedure table. `fivenum(x)` uses Tukey's hinges, which match the course method when n is even but not always when n is odd. In a **written** exam always use the procedure tables. In the lab, report what R gives. Percentile rank has no single R function: `(sum(x < 78) + 0.5)/length(x)*100` reproduces the formula.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · **03 · Lesson 3-3: Measures of Position** · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · 07 · Full worked answers